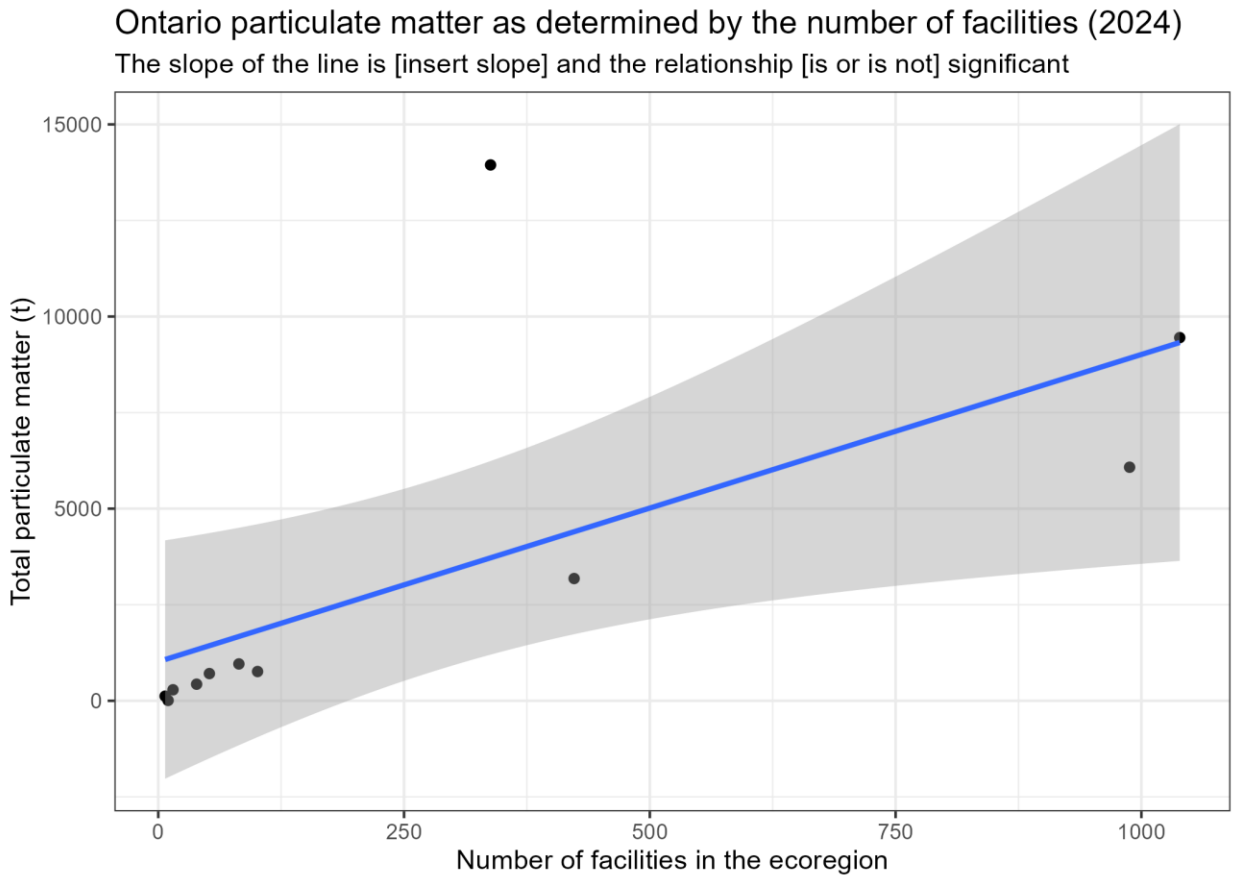
Practice exam: Particulate matter in Canada



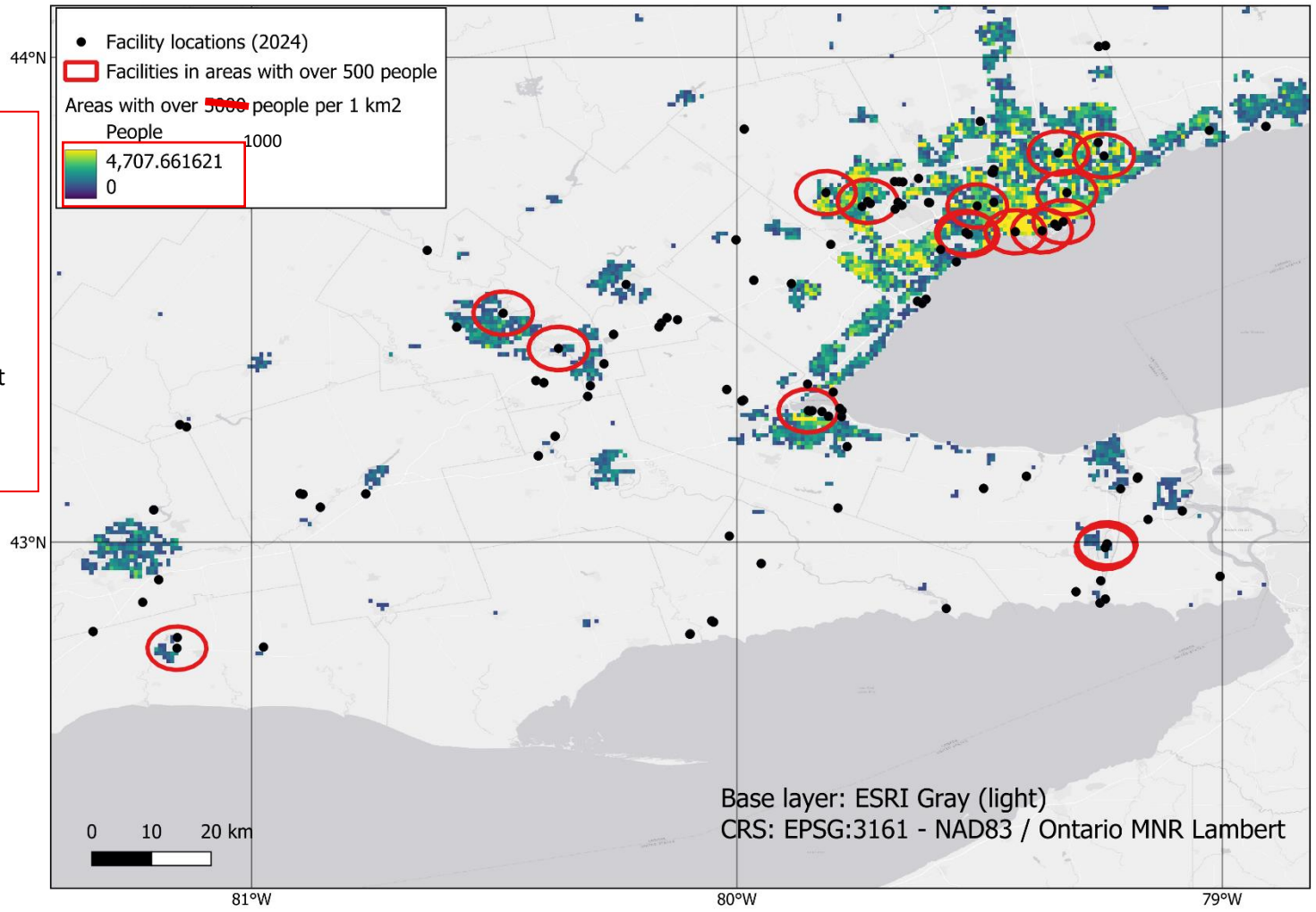
Datasets to work with (in data folder):

Dataset	Description
particulate_matter_ON.csv	Particulate matter releases from Ontario facilities (2020-2024)
ELC shapefile	The Ecological Land Classification (ELC) layers divide the landscape into ecological regions (ecoregions) based on features such as climate, geology, soils and vegetation.
can_ppp_2020_1km_Aggregated	This raster shows the estimated population of Canada in 2020, aggregated to a grid at approximately 1 km resolution. Each cell contains the estimated number of people living within that cell, rather than population density.

1. Please recreate the figure below. As part of the question, please add the slope and indicate whether or not the slope is significantly different from zero in the subtitle.



2. Please recreate the figure below. Red rings represent 5 km distance from facilities that are located in areas with over 500 people. Grid cells with over ~~5000~~ 1000 people are also shown on the map.



These range values on the raster symbology may not be aligned with the full range of the dataset, but by default when you change symbology it should show up like this

3. Recreate the figure below. Pay careful attention to the details in the figure title and subtitle.

